

Sylow Plan

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Definitions

- G be a finite group, where $|G| = p^n m$ and $\gcd(p, m) = 1$
- p is a prime
- $\text{Syl}_p(G)$ is the set of Sylow p -subgroups
- $X = \{S \subseteq G : |S| = p^n\}$

Steps of Proof

Number Theoretic Section

1. From def of X to $|X| = \binom{p^n m}{p^n}$
2. $|X| \equiv m \pmod{p}$ by Lemma 3.3(ii)
3. $p \nmid |X|$

Orbit Section

4. Left multiplication is a group action
5. For each S , there exists $T \in \text{Orb}_G(S)$ such that $1_G \in T$
6. $\text{Orb}_G(S) = \text{Orb}_G(T)$ by Corollary 2.17(i)
7. List the distinct orbits in X $\{\text{Orb}_G(S_i) : i \in [1, t]\}$, assume $1_G \in S_i$
8. Orbits partition by Corollary 2.17(ii)
9. If $V_1, V_2, \dots, V_n$ are disjoint finite sets, then the disjoint union $V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_n$ has $|V| = \sum_{i=1}^n |V_i|$
10. Rearrange/Assume $\{S_i\}$ is such that $p \nmid |\text{Orb}_G(S_i)|$ for $1 \leq i \leq r$ and $p \mid |\text{Orb}_G(S_i)|$ for $r < i \leq t$ for some arbitrary r .

Claim 1

11. If $g \in \text{Stab}_G(S_i)$ then $gS_i = S_i$ by definition
12. By assumption $1_G \in S_i$ then we have $g = g1_G \in gS_i$ Thus $\text{Stab}_G(S_i) \subseteq S_i$
13. By assumption $p \nmid |\text{Orb}_G(S_i)|$, also by Orbit-Stabiliser Theorem $|G| = |\text{Orb}_G(S_i)||\text{Stab}_G(S_i)|$
14. therefore $p^n \mid |\text{Stab}_G(S_i)|$
15. $|S_i| = p^n \leq |\text{Stab}_G(S_i)|$
16. $S_i \subseteq \text{Stab}_G(S_i)$ - - isn't necessary to state by the argument in 17
17. $S_i = \text{Stab}_G(S_i)$ - - by set theory (if $A \subseteq B, |A| \geq |B|$ then $A = B$)

Claim 2

18. Define a map $f : \{\text{Orb}_G(S_i) : 1 \leq i \leq r\} \rightarrow \text{Syl}_p(G)$ by $f(\text{Orb}_G(S_i)) = S_i$, this is well defined by Claim 1.
19. Let $1 \leq i, j \leq r$, and suppose that $S_i = S_j$. Then $\text{Orb}_G(S_i) = \text{Orb}_G(S_j)$
20. f is one to one
21. Let $P \in \text{Syl}_p(G)$, Then $P \in X$
22. $\text{Stab}_G(P) = P$ by Lemma 1.15
23. $p \nmid m = |G : P| = |G : \text{Stab}_G(P)| = |\text{Orb}_G(P)|$
24. $\text{Orb}_G(P) = \text{Orb}_G(S_i)$ for some $1 \leq i \leq r$
25. $P \in \text{Orb}_G(S_i)$
26. $P = gS_i$ for some $g \in G$
27. $1_G \in gS_i$
28. $g^{-1} \in S_i$
29. By Claim 1, $g \in S_i$
30. $gS_i = S_i$
31. $P = S_i = f(\text{Orb}_G(S_i))$ so f is onto
32. f is a bijection
33. $r = |\text{Syl}_p(G)|$

Conclusion

- 34. By Lemma, 3.3(ii), $|X| \equiv m \pmod{p}$
- 35. $|X| \equiv \sum_{i=1}^r |Orb_G(S_i)| \pmod{p}$
- 36. By Claim 1, $|Orb_G(S_i)| = |G : Stab_G(S_i)| = \frac{|G|}{|S_i|} = m$
- 37. By Claim 2, $r = |Syl_p(G)|$
- 38. $m \equiv mSyl_p(G) \pmod{p}$
- 39. As $p \nmid m$, m has an inverse u modulo p . Multiply by u .